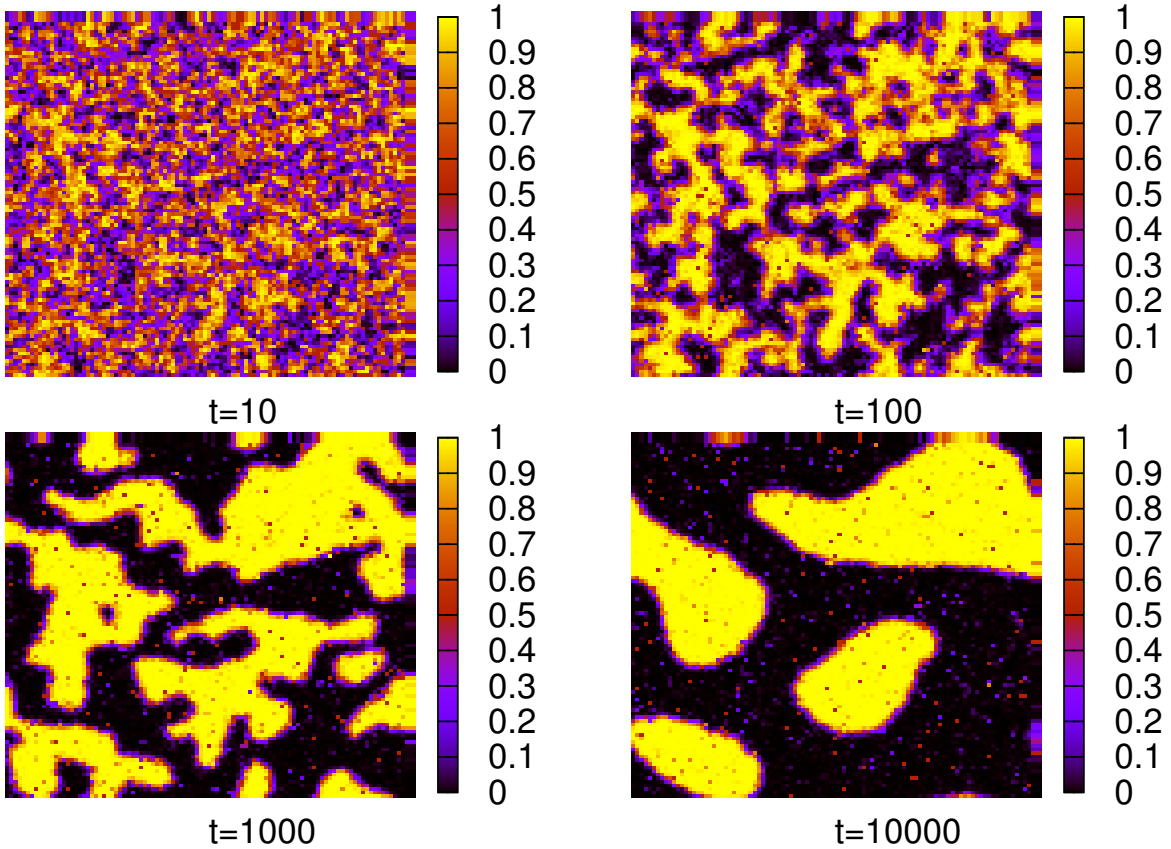

Fig 10. Spatial distribution of s_1, the probability that an agent will select word W_1 (Eq (1)). Results for a single-object model on a square lattice with $N = 10^2 \cdot 10^2 = 10^4$, $\alpha = 2$, $N_w = 2$, and with the renewal probability $p = 10^{-4}$. In this case, contrary to Fig 3, clusters of agents with the same dominant word grow steadily.

to this quantity come mainly from interfacial agents. Provided that the characteristic cluster size is l, we easily find a scaling relation $1 - m_L \sim l^{-1}$ [31]. From the time dependence of $1 - m_L$ (Fig 11), we conclude that $l \sim t^{0.41}$ and such a value seems to be independent of the renewal probability $p > 0$. Only for $p = 0$, we obtain a much slower increase of l, perhaps logarithmic (in time), which reflects the trapping of the model in a disordered configuration as, for example, in Fig 3. A small deviation from the Ising model increase $l \sim t^{1/2}$ [20] may be attributed perhaps to a diffusive structure of an interface in our model. Let us notice that a similar increase $l \sim t^{0.45}$ was observed also in certain opinion-formation models that are expected to have a dynamics with an effective surface-tension [32].

We also analysed how the coarsening dynamics depends on α. Our results for the renewal probability $p = 10^{-3}$ are shown in Fig 12. It seems that the asymptotic decay of $1 - m_L$ is characterized by the same exponent for any $\alpha > 1$. A qualitatively different behavior is seen only for $\alpha = 1$, where the model does not coarsen. These results suggest that the behavior of our model, also with respect to other properties, should not depend on a particular choice of α as long as $\alpha > 1$ (superlinear reinforcement).

The population renewal affects also a multi-object version of our model. In the absence of the renewal, the model on the complete graph (for $N_o = 10$ and $N_w = 30$) develops a language with a reduced success rate (Fig 8) and with no clear object-word mapping (Fig 9). However, even a tiny renewal probability ($p = 10^{-4}$) considerably increases the success rate of communication (Fig 8). We do not present our numerical data here but in this case, a clear object-word mapping does emerge, similar to that in the upper panel of Fig 9. Also a multi-object square-lattice version of the model behaves differently for

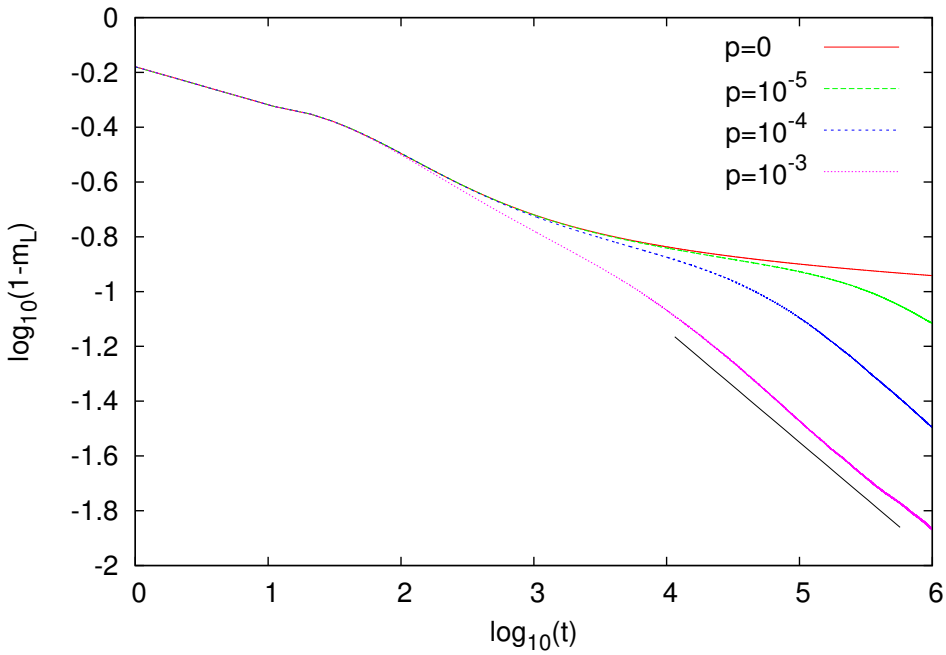

Fig 11. Time dependence of $1 - m_L$ for several values of the renewal probability p. Results for a single-object, square-lattice version of our model. Simulations were made for $\alpha = 2$, $N_w = 2$, and $N = 200 \cdot 200 = 4 \cdot 10^4$, and the results are averages of 20 independent runs. The line segment has a slope corresponding to $t^{-0.41}$.

$p > 0$. Indeed, the snapshot configurations (Fig 7) show that in this case the model does not get trapped (as for $p = 0$) but coarsens similarly to the single-object version with population renewal (Fig 10).

The overall behavior of our models is summarized in Table 1.

Discussion and conclusions

The objective of the present study was to examine the emergence of linguistic conventions in a multi-agent model with reinforcement learning. Models of this kind have a potential to generate a complex language, which reflects their multi-object and multi-agent structure, but global aspects of their dynamics are rather purely understood. This is much in contrast to some simpler models with agreement dynamics, like Ising model or Naming Game, which due to the surface tension have curvature-driven dynamics [20], or voter model, which lacks the surface tension and has a much different dynamics [33].

In the single-object version, agents do not need to recognize the object and each communication attempt is in a sense successful. It turns out that it is the structure of the network of interactions between agents that plays a decisive role and determines the asymptotic state of the model. While for a complete graph or sufficiently dense random graphs, a global consensus is reached and all agents use the same word for communication, on finite-dimensional lattices or sparse random graphs, only a local consensus is reached and the model gets trapped in a disordered configuration.

In the Naming Game, which is an alternative model describing formation of linguistic consensus, agents also try to establish a name for a given object. However, in this case a global consensus is much easier to reach except on networks with a strong community structure [34, 35]. Such a strong tendency to reach a consensus could be explained using the notion of an effective surface tension, which is generated in the Naming Game. It turns out that when a population renewal is introduced, the surface

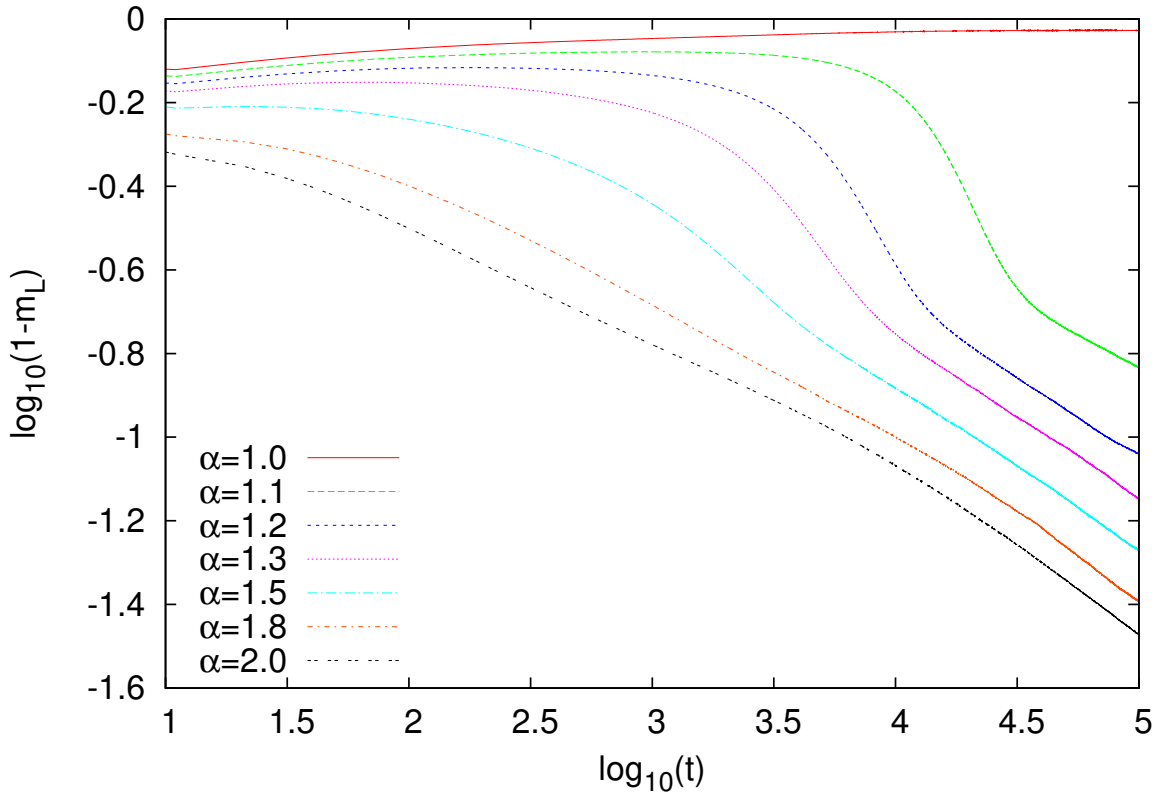

Fig 12. Time dependence of $1 - m_L$ for several values of α. Results for a single-object square-lattice version of our model for the renewal probability $p = 10^{-3}$. Simulations were made for $N_w = 2$ and $N = 200 \cdot 200 = 4 \cdot 10^4$, and the results are averages of 20 independent runs.

tension emerges also in our model and its evolution toward consensus is much enhanced. However, such a curvature-driven evolution appears only for $\alpha > 1$, which indicates the importance of a superlinear reinforcement. Our study thus shows that a physical intuition developed for some statistical-mechanics models may be also used to understand to some extent reinforcement learning systems.

Having in mind linguistic contexts, the multi-object version of our model is more interesting. The results show that in this case the structure of the network plays an important role as well. On complete graphs, an efficient global communication may be established, such that all agents unambiguously match each object with the same corresponding word. On finite-dimensional lattices, such a mapping is again only local (and partial) and the model gets trapped in a disordered configuration. In the multi-object version, in addition to the network structure, other parameters are also important, namely the number of objects N_o and the number of words that agents have at their disposal N_w. Our simulations suggest that a unique object-word matching may emerge only when N_w is considerably greater than N_o. If this is not the case, the resulting communication is less efficient and the emerging language contains some homonyms and synonyms. Of course, such behavior should be by no means considered as undesirable or unrealistic, since all natural languages contain such forms. Further studies concerning, for example, the frequency and durability of homonyms and synonyms in our model would be desirable, but are left for the future.

In the multi-object version, the population renewal also enhances formation of an efficient communication. The population renewal basically resets the weights of an agent, and thus it plays a role similar to forgetting, which is a factor already known to improve the performance of reinforcement learning systems [36]. Our snapshot configurations show that also in this case, the population renewal most